

to the increase in internal energy.

14. **Ans: C**

Using $v_0 = \omega x_0$, $3 = \frac{2p}{12} x_0$ giving $x_0 = 5.730$ cm

Using $|a_0| = \omega^2 x_0$, $a_0 = \left(\frac{2p}{12}\right)^2 \times 5.730 = 1.571$ cm s⁻¹

Hence maximum restoring force = $5.0 \times 1.571 \times 10^{-2} = 7.9 \times 10^{-2}$ N